Household Electricity Forecasting

Assignment 1 | Project proposal | SDG 7 | Group 7

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1. Problem, users and SDG contribution

We propose an AI-based tool that helps households with access to time-stamped electricity records anticipate the next day's hourly consumption and identify expected high-consumption periods. The practical problem is turning historical meter readings into an understandable forecast for daily energy planning.

The project aligns with SDG 7, particularly Target 7.3, which calls for doubling the global rate of improvement in energy efficiency by 2030 [1]. Our intended contribution is better consumption awareness and planning. Forecast accuracy is the initial success measure; actual energy savings would require a separate user study and are not claimed here.

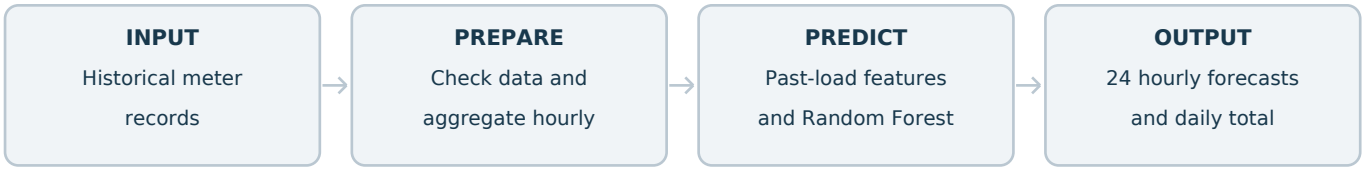
2. Dataset and preparation

We use UCI's Individual Household Electric Power Consumption dataset: 2,075,259 minute-level records from one home in Sceaux, France, from December 2006 to November 2010 [2]. It provides historical load patterns for a proof of concept; it does not establish performance across households.

Our checks found no invalid or duplicate timestamps, absent minute timestamps, or negative measurements. Each measurement column contained 25,979 missing values (1.25%) at the same timestamps, forming 71 gaps, including six longer than 24 hours. We did not impute missing readings.

Hourly energy (kWh) was calculated as the sum of 60 minute-average active power measurements (kW), divided by 60. Of 34,589 hourly intervals, 34,085 were complete. The remaining 504 intervals were retained with missing energy values to preserve the time grid. Lag features were created before filtering; requiring complete features and 24 usable hours per day left 1,080 days (25,920 hours).

3. Proposed input and output flow



Planned interface: upload consumption history, view a next-day hourly chart and daily total, and inspect expected high-consumption hours. The current milestone is an offline forecasting experiment; the user interface will be developed in Assignment 2.

4. Model choice and first experiment

Random Forest regression was selected as an initial model for nonlinear relationships between calendar information and past consumption. Inputs are hour, weekday, month, weekend status, consumption at the same hour 1, 2, 3 and 7 days earlier, and the previous complete day's mean and maximum hourly consumption. Quality flags and future actual consumption are excluded from the inputs.

Configuration: 200 trees, maximum depth 15, minimum leaf size 5, random seed 42. The model was fitted once on the earliest 864 eligible days (20,736 hours; 25 Dec 2006–20 Feb 2010) and tested on the latest 216 eligible days (5,184 hours; 22 Feb–25 Nov 2010). These are selected complete days, not uninterrupted calendar periods. Each test day is forecast at its start using only measurements available by then, including earlier test days. No test-based parameter tuning was performed in this first experiment.

Two reference methods predict consumption using the same hour of the previous day or previous week. All methods were evaluated on identical test hours using mean absolute error (MAE) and root mean squared error (RMSE).

Method	MAE (kWh)	RMSE (kWh)
Previous day, same hour	0.5134	0.7662
Previous week, same hour	0.5546	0.8047
Random Forest	0.4205	0.5827

Random Forest reduced MAE by 18.1% and RMSE by 24.0% versus the stronger, previous-day reference. These percentages describe error reduction, not energy savings or a classification accuracy score.

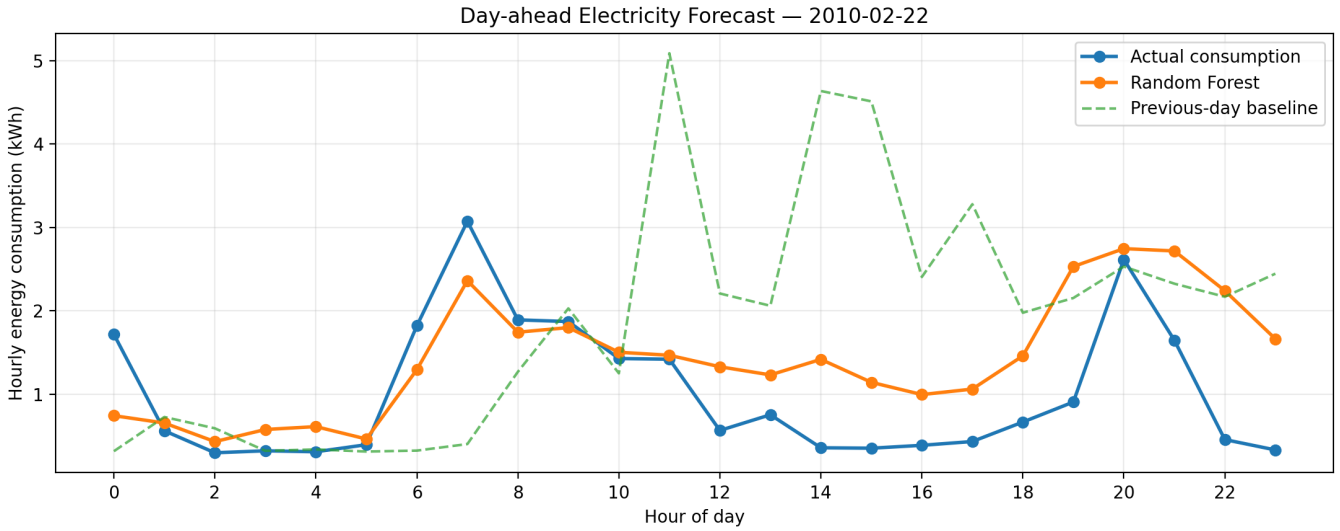


Figure 1. First test day, selected without searching for a favorable example. Morning patterns are partly captured, but afternoon and late-evening consumption are overestimated. Overall performance is based on all 5,184 test hours.

5. Limitations and next steps

The data cover one historical household; performance on other homes is untested. Complete-day filtering may bias evaluation toward well-observed periods. Weather, occupancy and appliance schedules are unavailable. High consumption is not automatically waste. The tool will label outputs as estimates and avoid guaranteed savings claims; future uploads should be handled privately because load histories can reveal routines. Next steps are chronological validation within the training period, broader error analysis and a simple working interface.

References

[1] United Nations DESA. Goal 7, Target 7.3. https://sdgs.un.org/goals/goal7
[2] Hebrail, G. & Berard, A. (2006). Individual Household Electric Power Consumption. UCI Machine Learning Repository. https://doi.org/10.24432/C58K54. CC BY 4.0.